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Rendering SSSOM ontology mappings as RDF named graphs and RDF 1.2 triple terms

Andra Waagmeester ¹, Jerven Bolleman ², Ruben Taelman ³, and Thomas Pellissier Tanon ⁴

¹ Amsterdam UMC ² SIB Swiss Institute of Bioinformatics  ³ IDLab, Ghent University – imec 

⁴ Independent

Introduction

The Simple Standard for Sharing Ontological Mappings (SSSOM) (Matentzoglou et al., 2022) is widely used in the life sciences to share mappings between terms of different datasets and ontologies. A SSSOM file is a hybrid artefact: a YAML metadata header (carried in #-prefixed comment lines) followed by a tab-separated (TSV) table of mappings. Although the standard carries ontological commitments, the files are most often consumed as plain dataframes.

SSSOM already defines an RDF/OWL representation that describes a *mapping set as a whole*. During the GOBLIN hackathon, we explored two alternative, complementary RDF renderings that instead make the individual *mapping* the first-class object, and we compared the design trade-offs between them. This report documents that exploration. **It is an early, first-draft proposal intended to start a conversation rather than a finished specification** (see [Discussion](#)).

Approach

The native SSSOM format

The core of each TSV row — `subject_id`, `predicate_id`, `object_id` — is an RDF triple; the remaining columns (`mapping_justification`, `labels`, ...) and the header (`provenance`, `confidence`, the `curie_map`) are metadata about that triple or about the set. A mapping is therefore, ontologically, *a statement about a statement*, which can be expressed in RDF at two different layers.

Two renderings

We built a single streaming serializer that emits both representations from the SSSOM TSV files. **For transparency: the serializer was generated by prompting a large language model (Claude); it is a starting point to be audited, not a reference implementation.** It parses the YAML header, expands every CURIE to a full IRI using the file's authoritative `curie_map`, and writes RDF.

Named graphs (TriG). Each mapping set becomes one named graph (graph IRI = `mapping_set_id`); the core triples live inside the graph and set-level provenance is asserted about the graph IRI in the default graph. This sits at the RDF *dataset* layer and faithfully models `sssom:MappingSet`. It is universally supported (plain SPARQL 1.1 GRAPH), but per-mapping metadata is not attached.

RDF 1.2 (Turtle 1.2). Each mapping triple is asserted and annotated with an RDF 1.2 triple term (W3C RDF-star Working Group, 2025), using the `{ | ... | }` annotation syntax that desugars to a reifier (`_:m rdf:reifies <<( s p o )>>`) carrying the justification, labels

and a back-link to the set. This sits at the *abstract-syntax* layer, models `sssom:Mapping`, and is lossless — at the cost of a ~5x statement blow-up and a dependency on an RDF 1.2-aware engine.

CURIE expansion and IRI-unsafe local names

The header CURIE map is treated as authoritative. However, some local-name parts are not valid inside an IRI — the canonical case is the SMILES: “prefix”, which encodes chemical structures rather than identifiers and contains characters (`/`, `#`, `[`, `]`, `=`, ...) that IRI syntax forbids. We percent-encode the local part (per RFC 3987), leaving the namespace literal, so the result is a syntactically valid IRI.

Sources

We used two sources. The EBI Ontology Lookup Service (OLS) (Côté et al., 2006), which publishes per-ontology SSSOM extracts, was processed in full. We also began extracting ontology cross-references from Wikidata as SSSOM; this was not completed within the hackathon and is reported as future work.

Results

The full set of OLS SSSOM extracts — **271 mapping sets, 6,238,000 mappings** — was rendered into both formats in ~91 s. Both outputs were validated by parsing with a strict RDF 1.2 parser (pyoxigraph). The renderings, schema diagrams (drawn in ShEx and, for RDF 1.2, in ShEx *style* since ShEx 2.1 has no triple-term construct yet), and tooling notes are published in the project code repository (https://github.com/andrawaag/GOBLIN_Oviedo_hackathon_2026); the gzipped RDF is distributed there via Git LFS. The two renderings are loaded into Oxigraph (Pellissier Tanon, 2024), which serves a SPARQL endpoint, and queried from a browser with Comunica (Taelman et al., 2018), which can additionally federate the local endpoint with remote endpoints such as Wikidata.

Table 1: Summary of the two RDF renderings of the OLS SSSOM extracts.

Property	Named graphs (TriG)	RDF 1.2 (Turtle 1.2)
RDF layer	dataset / context (quads)	abstract syntax (triple terms)
Granularity	one graph per mapping set	per mapping (<code>{\ ... \ }</code>)
Models	<code>sssom:MappingSet</code>	<code>sssom:Mapping</code>
Metadata carried	set-level provenance	justification + labels (lossless)
Tooling	universal (SPARQL 1.1)	needs SPARQL 1.2 engine
Size	1.5 GB (~6.24M quads)	3.7 GB (~34M statements)

Discussion

The two renderings are not competing encodings of the same idea: they attach metadata at different layers of the RDF stack and model different SSSOM entities (named graph $\leftrightarrow$ MappingSet, triple term $\leftrightarrow$ Mapping). They therefore *compose* — the named-graph file is the portable, partition-by-source view, and the RDF 1.2 file is the lossless, per-mapping master.

Relative to the existing SSSOM/RDF representation, our approach makes two deliberate scope choices. First, we do not support mappings between literals (e.g. `"hello"@en skos:closeMatch "hallo"@nl`), which require generalized RDF that RDF 1.2 triple terms do not provide; as most datasets and tooling do not use generalized RDF, we accept that this excludes a small



number of files. Second, we describe each mapping independently rather than describing the file as a whole.

This work is explicitly a **first-draft proposal** and the outputs require validation before reuse:

- **IRIs must be minted.** The serializer emits two placeholder IRIs under w3id.org that we do not own: a convenience predicate `https://w3id.org/sssom/mapping_set` (not a standard SSSOM slot) and a `https://w3id.org/sssom/.well-known/curie/...` fallback for unresolved CURIEs. The reused sssom: slot IRIs are assumed to match the official SSSOM model and must be verified.
- **Modelling is not yet validated.** Parsing with an RDF 1.2 parser confirms syntactic validity, not correct modelling; the shapes need review against the SSSOM community.
- **Propagatable slots are not yet implemented.** Header slots such as `license` and `mapping_date`, and `extension_definitions`, are part of the proposed approach but not yet emitted by the script.

Future work includes completing the Wikidata-to-SSSOM extraction and federating it with the OLS mappings through Comunica, aligning the vocabulary with the official SSSOM/RDF model, minting the required IRIs, and validating the RDF 1.2 shapes once ShEx (or an equivalent) gains triple-term support.

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Author contributions

Andra Waagmeester: Conceptualization, Software, Writing – original draft Jerven Bolleman: Conceptualization, Writing – review & editing Ruben Taelman: Software, Writing – review & editing Thomas Pellissier Tanon: Conceptualization, Software